# Mukesh Vemulapalli

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## Education

**Rajiv Gandhi University of Knowledge Technologies** — Nuzvid, India
*Bachelor of Technology in Electronics, and Communications Engineering* — *Sep 2022 – May 2026*

## Experience

**AI Engineer (Intern)** — Aug 2025 – Mar 2026
*Blotout* — *Fremont, California, United States (Remote)*

- Built and optimized AI pipelines for EvoAI, powering hyper-personalized product recommendations, and dynamic product ranking.
- Developed agentic workflows powering real-time order tracking, delayed-shipment alerts, automated ticket creation, and live-agent handoff.
- Implemented AI-driven personalization systems for contextual pricing, promo-code recommendations, and behavior-based product suggestions.
- Improved data ingestion and feature-engineering pipelines to ensure clean, scalable data flow for recommendation engines and LLM-powered shopping agents.

**AI Engineer (Intern)** — Jan 2025 – Jul 2025
*Amberley Tech* — *Remote, India*

- Built and deployed Bittensor subnets including a Whisper-based audio caption subnet with miner-validator architecture, while contributing as a miner across multiple decentralized AI subnets to optimize AI-blockchain integration.
- Developed advanced LLM applications by fine-tuning DeBERTa for AI vs. human text classification, achieving an F1 score of 99.71% and an ROC AUC of 99.96% in detecting AI-generated content under adversarial conditions.
- Created an AI content-generation multi-agent system that produces long-form blog articles and automatically publishes them to the Arweave decentralized storage blockchain.
- Engineered an Ethereum Model Context Protocol (MCP) server in Python, enabling real-time AI agent interaction with blockchain data and intelligent querying of on-chain information.

**AI Intern** — Aug 2024 – Dec 2024
*Cloud Counselage Pvt. Ltd.* — *Remote, India*

- Developed an AI Code Plagiarism Detector for the IAC (Industry-Academia Community) to review projects, reducing evaluation time by 90% and deploying it into production.
- Worked on a Plant Disease Detection system using Deep Learning.
- Designed a Chatbot using Deep Learning for diverse applications.

**LLM Engineer** — Jul 2024
*Sentient Matters* — *Contract, Remote, India*

- Developed a banking helpline bot using Llama-2 and fine-tuned the model for banking purposes.
- Reduced human intervention by automating customer support tasks and deployed the solution to production.

## Projects

**LLM Trading Agent** | *Python, LLMs, NumPy, Numba* | GitHub Link — Jul 2025 – Aug 2025

- Architected an autonomous AI trading system leveraging Large Language Models (LLMs) and Chain-of-Thought (CoT) reasoning to execute real-time cryptocurrency market analysis, integrating sentiment analysis (Fear & Greed Index) with quantitative data to generate high-confidence trade signals.
- Engineered a high-frequency technical analysis engine using Python, NumPy, and Numba JIT compilation, optimizing the calculation of 25+ complex indicators (RSI, MACD, VWAP, Hurst Exponent) to achieve low-latency signal processing on streaming OHLCV data.
- Developed a robust, asynchronous infrastructure using asyncio, aiohttp, and ccxt, featuring automated risk management (dynamic Stop-Loss/Take-Profit), fault-tolerant model fallback mechanisms, and a Streamlit dashboard for visualizing trade performance and P&L metrics.

**High-Performance DeBERTa Fine-Tuning** | *PyTorch, DeBERTa, NLP* | GitHub Link Jun 2025

- Engineered a state-of-the-art NLP pipeline for Human vs. AI text classification by fine-tuning Microsoft's DeBERTa-v3-large, developing a custom PyTorch 'AdvancedTrainer' class to handle complex training loops and data preprocessing.
- Optimized training throughput and memory efficiency by implementing Multi-GPU distributed training ('DataParallel'), Mixed-Precision (FP16) acceleration, and gradient accumulation, enabling the efficient processing of large-scale datasets on constrained hardware.
- Achieved superior model performance with a 0.9971 F1-score, 0.9996 ROC-AUC, and 0.9993 Average Precision, designing a comprehensive evaluation framework that tracks false-positive rates to ensure high-reliability predictions.

**AI-Powered Inventory Management System** | *Gemini Pro, Streamlit, SQLite* | GitHub Link Feb 2024

- Developed a Generative AI-driven analytics platform using Google Gemini Pro and PandasAI, enabling non-technical users to query inventory databases via Natural Language Processing (NLP) and automatically converting text to executable SQL.
- Engineered a robust Streamlit dashboard integrating SQLite and Plotly, featuring dynamic data visualization, automated stock prediction algorithms, and bulk data processing capabilities through Excel integration.
- Implemented intelligent reporting agents to automate business intelligence tasks, utilizing LLM reasoning to generate real-time insights on sales trends, stock depletion risks, and product categorization.

## Technical Skills

**Technologies**: Generative AI, LLMs, AI Agents, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing
**Languages**: Python, C, C++, Dart, JavaScript, Rust
**Frameworks**: TensorFlow, PyTorch, Keras, LangChain, OpenAI, LlamaIndex, Langfuse, FastAPI, Flask, Django, Agno, Google's Agents SDK, CrewAI, LangGraph, Agent2Agent Protocol
**Libraries**: NumPy, Pandas, Matplotlib, OpenCV, MediaPipe, Scikit-learn, NLTK
**Tools**: VS Code, Postman, Git, Docker
**Databases**: MySQL, MongoDB, Pinecone
**Cloud Platforms**: AWS, Google Cloud
**Hardware**: Arduino, NXP i.MX RT117H Board

## Achievements

Top 5 in VLSI Design Contest at VLSI Design Conference 2025 for developing a Museum Guide Bot with gesture recognition and voice recognition using the NXP i.MX RT117H board.

Finalist in Avishkaar Season 2 for developing an AI Travel Assistant.

Top 10 in the Regional Finale of NXP AIM (Artificial Intelligence in Mobility) for building an autonomous car in Gazebo and ROS using computer vision techniques for path planning.

Participated in Hacktoberfest 2024, contributing to an AI Medical Chatbot and various ML projects such as a movie recommendation system, JEE rank analysis and institute prediction system.